CASE B · THE DATA MOAT

The Financial Model, Explained

Every tab, every term and how it all connects to the case brief, in plain English for a first-time reader.

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Who this is for

Anyone opening our workbook for the first time: a teammate building the slides, a mentor, or a judge. No finance background needed.

The honest headline, in three numbers (Base case, model output)

4.0% of dropouts prevented (the PS assumed 25%)	**0.55x** owner's return on hard benefits, fees and staff time (1.7x with admissions)	**3.6x** lifetime value ÷ cost to win a customer (the PS said 10.7x)

What is inside

13 tabs explained, one by one	**50** jargon terms in plain words	**PS** linked to the case brief throughout
5 min to try it yourself	**102** inputs, each with a source	**1** honest page on limits

START HERE

How to read this guide

This guide walks you through our financial model (an Excel workbook) the way a friend would: what each tab is for, what the jargon means, and where the case brief asks for it. The case brief is called the **PS** (problem statement) throughout. New words are explained as they appear, and again in Part 2.

IN PLAIN ENGLISH

The whole model in one breath. We ask three questions. (1) Is our software worth its price to a coaching institute? (2) Do we earn more from an institute than it costs us to win and serve it? (3) What does that add up to over three years, and how much money do we need? Every tab in the workbook feeds one of those three answers.

FROM THE PS (the case brief)

Where this sits in the competition. The PS gives 30% of the marks to the edtech market and business (Section 4). This model is its heart: *Revenue model & unit economics* (20%, including the **3-year projection**) and part of *Market & go-to-market* (10%) (Section 10). Teams that state their assumptions and limitations score higher than teams that oversell, so we show ranges and warnings.

In plain English the idea without jargon	**From the PS** where the case brief asks for it	**Worked example** real numbers from the model	**Watch out** limits and soft spots

About the numbers. Every number is the **Base** scenario of model v1.1 CFO audit · 2026-10-22 unless the text says Bear or Bull. They are model outputs built on sourced and assumed inputs, not measurements. **Short on time?** Read Part 1, then the Cover, Institute_ROI, Unit_Econ and Projection tabs, then Part 6.

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PART 1

The big picture

Before the tabs, the story: what questions the model answers, why the PS's own example is too good to be true, and how a financial model is put together.

1.1 The questions behind the model

Think of a gym. The owner of a gym asks *"will members feel the fee is worth it?"* The company that built the gym's app asks *"after support staff and advertising, do we make money from each gym, and how big can this get?"* Our case is the same, with coaching institutes instead of gyms.

- **Question 1, the institute's view (tab Institute_ROI).** For an institute with 400 students: how many students can our software help keep, how much fee does that save, and is it more than the software costs?
- **Question 2, our view (tab Unit_Econ).** For each institute we sign: how much do we earn a month, what did it cost to win, how long does it stay, and do we get our money back?
- **Question 3, the business over three years (tab Projection).** How many institutes can we realistically win, what do we earn and spend each month, and how much funding do we need?

> **FROM THE PS (the case brief)**
> These are PS Section 3, questions **B1** ("Who is the buyer … and what is the measurable ROI for an institute?") and **B2** ("What do you charge … and what are CAC, payback, churn and LTV?"), plus the *3-year projection* named in the judging rubric (Sections 4 and 10). Don't worry about the abbreviations yet; Part 2 explains each one.

1.2 The PS's own example, and why it is too good to be true

PS Section 6 gives a deliberately simple worked example. Here it is, step by step:

Step	PS calculation	Result
Fees lost to dropouts	400 students × 12% × ₹36,000	₹17,28,000
Saved if the platform prevents 25%	12 students × ₹36,000	₹4,32,000
Platform cost at ₹40 a student a month	400 × ₹40 × 12	₹1,92,000
Return on platform cost	₹4,32,000 ÷ ₹1,92,000	about 2.25x
Monthly contribution per institute	₹16,000 revenue − ₹5,400 costs	₹10,600
Payback and lifetime value	₹33,000 ÷ ₹10,600; ₹10,600 × ~33 months	3.1 months; LTV/CAC about 10x

The PS then says, in its own words, **"Your job is to challenge this."** The numbers look so attractive that judges will question them. Our workbook keeps the PS example as a column called *Guide (naive)* next to our own numbers and shows what changes once real-world friction is added:

1. Not every dropout can be prevented. A quarter leave suddenly, and some leave before we have any data on them.
2. Our early-warning model is good, not magic: at a workload a mentor can handle it finds about a third of dropouts, and most of its flags are false alarms (the technical team measured this).
3. Not every rupee of fee is lost: instalment payers simply stop paying, and upfront payers may get a refund.
4. Classrooms empty during exams, so a bill based on active students shrinks at certain times of year.
5. Customers leave. A monthly churn of 3% sounds small, but it adds up and spikes when the new session starts.
6. Winning a customer costs more than a salesperson's time: free pilots that do not convert, onboarding, sales overhead.

Here is the effect on the headline numbers, for the same 400-student institute:

Headline	PS example	Our model (Base)
Share of dropouts prevented	25%	4.0%
Students kept per year	12 of 48	1.9 of 48
Institute return, fees only	2.25x	0.35x
Institute return, fees + staff time (hard benefits)	2.25x	0.55x
Institute return, all benefits incl. 1% more admissions	2.25x	1.7x
LTV ÷ CAC	10.7x (undiscounted)	3.6x discounted, 4.8x undiscounted
Payback	3.1 months	5.7 months (simple)

WATCH OUT

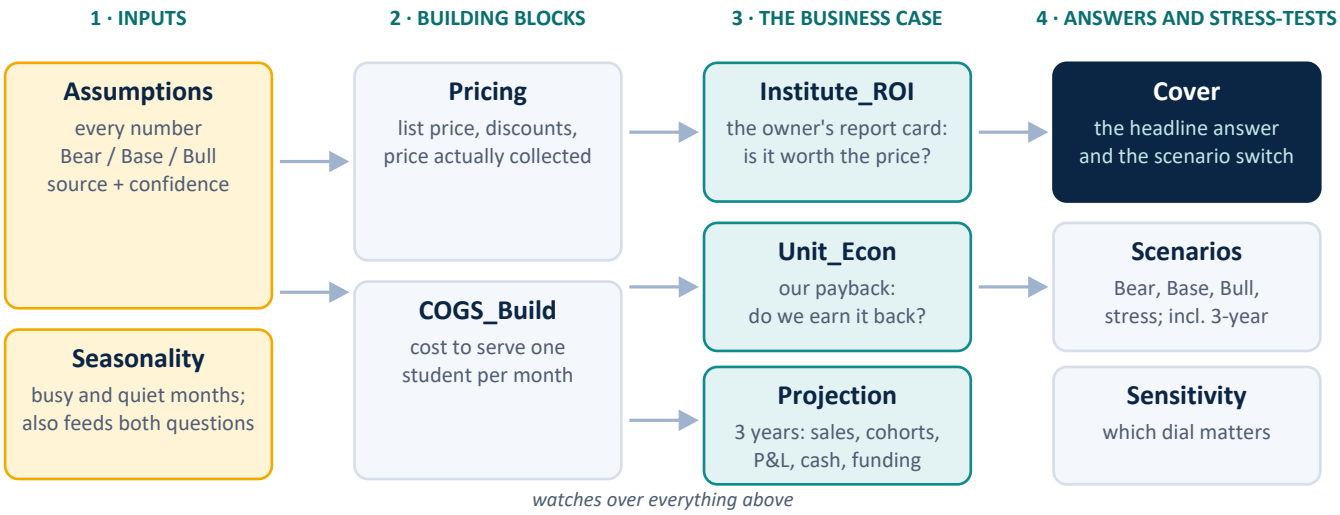
Our numbers are lower than the PS's on every line. The one that matters most is the owner's return: counting only what they can bank (fees and staff time), the product returns **0.55x** of its cost, not 2.25x. It reaches 1.7x only if better results bring 1% more admissions, which nobody has proved yet. We show all three numbers on the Cover on purpose, and the Institute_ROI section explains why.

1.3 What a financial model really is

A financial model is a calculator you can ask *"what if?"* You type in your beliefs (the **assumptions**), the calculator turns them into money numbers, and you change a belief to see what happens. Its job is not to predict the future. Its job is to show **which beliefs matter** and **how bad things could get**.

Ours is built in six steps, and each step has its own tabs:

Step	What happens	Tabs
1 Write down the beliefs	Every input, with Bear / Base / Bull values, a source and a confidence label	Assumptions, Seasonality
2 Build the blocks	What price we really collect; what it costs to serve one student	Pricing, COGS_Build
3 Answer the two core questions	The institute's ROI; our own unit economics	Institute_ROI, Unit_Econ
4 Project three years	Sales, customers, P&L, cash and the funding we need	Projection
5 Stress-test	Scenarios side by side; which single input matters most	Scenarios, Sensitivity
6 Guard rails	Automatic checks and a source for every input	Checks, Sources



The workbook at a glance: inputs flow right, through building blocks, into the business case, and out as answers. The Cover tab is the front door.

Three house rules keep it trustworthy:

- **One home for numbers.** Every number lives on the Assumptions tab (plus the calendar on Seasonality). Calculation tabs only pull from there.
- **Formulas are never typed over.** Colour tells you what is what: **blue** is a typed input, black is a formula, **green** is a number pulled from another tab.
- **Ranges, not single points.** Each input has a Bear, Base and Bull value so you can see the spread.

FROM THE PS (the case brief)

PS Section 9, Round 1, deliverable 2: "Financial model with clearly listed assumptions and at least one sensitivity analysis… Formulas must be live, not hard-coded." Here is how we meet it, and what Round 2 adds:

PS requirement	How we meet it
Live formulas, not hard-coded	Every calculation is a formula. The only typed numbers are the blue inputs. A full recalculation shows 0 formula errors.
Clearly listed assumptions	102 inputs on Assumptions, each with unit, Bear / Base / Bull, type, confidence, source and a scenario-driver flag.
At least one sensitivity analysis	Sensitivity tab: a tornado of 17 inputs and two what-if grids.
Round 2: 3-year projection, scenarios, sensitivities	All three are in the workbook: Projection (36 months), Scenarios (Bear / Base / Bull plus a Stress test) and Sensitivity. A 3-year sensitivity grid is still to come.

PART 2

Jargon buster

Read this like a dictionary, not a chapter: 50 terms, grouped by the question they help answer. Examples use the Base case, a typical 400-student institute with a ₹75,000 annual fee.

A · About the institute's money

Term	In plain words	Where it shows up
Dropout	A student who leaves in the middle of the session, or stops paying. Base case: 12% of students a year, so 48 out of 400.	Assumptions → Institute
Upfront vs instalments	Some parents pay the whole fee at admission (upfront). Most pay in 2–4 parts (instalments). When a student leaves, only the *unpaid* instalments are really lost.	Base: 25% upfront; 3 instalments
Pro-rata refund	Money back for the part of the course not used: leave after 4 of 10 months and you get back 6/10 of what you paid. Ministry of Education guidelines say full payers should get this within 10 days, but they are advisory and many big chains do not follow them.	Assumed: 30% of upfront leavers actually refunded
Fee lost per dropout	The part of the fee the institute truly loses: unpaid instalments for instalment payers, a refund for upfront payers, depending on the month the student leaves.	Base: ₹29,799 on average
Active student	A student currently in class. Our bill is per *active* student, so it shrinks when classes empty out.	Seasonality
Seasonality	The yearly rhythm: admissions in April to June, thin classes in exam months, then the cycle restarts.	Seasonality tab
GST and input credit	GST is the 18% sales tax on our software. A GST-registered business can claim it back (*input tax credit*). A small unregistered one cannot, so for them it is a real extra cost.	Pricing; assumed 60% reclaim it

B · About how the model is built

Term	In plain words	Where it shows up
Assumption	A number we believe but have not measured. Each one carries Bear, Base and Bull values and a label for how much evidence stands behind it.	Assumptions
Bear / Base / Bull	Three versions of the key inputs. **Bear** = worse for our business, **Base** = our best guess, **Bull** = better. Only the 13 scenario drivers change; everything else stays at Base. Bear is not always the smaller number: high churn is Bear.	Cover switch
Scenario driver	One of the 13 inputs the scenario switch actually moves (price, trial effect, admissions gain, recall, precision, conversion, churn, market size and a few more). Marked Y in column N of Assumptions.	Assumptions col N
Stress test	Every input at its worst at once. It almost never happens, so it is a survival test, not a forecast. Shown as its own column on Scenarios.	Scenarios
Live formula vs hard-coded	A live formula recalculates when an input changes (for example =B2*B3). A hard-coded number is just typed in and goes stale. The PS requires live formulas.	all calculation tabs
Named range	A nickname for a cell, so a formula can read `=ListPrice` instead of `=Assumptions!G25`.	Assumptions
Scenario	One complete set of Bear, Base or Bull values. The switch on the Cover chooses which set the model shows.	Cover cell C6; Scenarios

Term	In plain words	Where it shows up
Sensitivity / tornado	Move *one* input from its Bear value to its Bull value, keep the rest at Base, and see how far the answer swings. Sorted longest bar first, the chart looks like a tornado.	Sensitivity
Evidence labels	**Researched** (found in a source), **Public** (public dataset), **Guide** (from the PS), **Assumption** (our judgement), **Structural** (a modelling convention).	Assumptions col H; Sources

C · About preventing dropouts

Term	In plain words	Where it shows up
Recall	Of the students who really will leave, the share our early warning flags at a workload a mentor can handle. Base: 32% for under-18s, 42% for adults, matching what the technical team measured.	Institute_ROI
Precision	Of the students we flag, the share who really would have left. 15% means about six of every seven flags are false alarms. It drives how much mentor time is spent.	Institute_ROI
Catchable share	Dropouts that happen after we have enough test data to notice. A student who leaves in the first 2 months is invisible to us. Base: 87% are catchable.	Institute_ROI
RCT	Randomised controlled trial: a fair experiment where some students get outreach and a similar group does not. We borrowed an average effect of 15% from 8 such trials in other settings.	Institute_ROI
Effective reduction	The share of *all* dropouts we actually prevent: catchable × recall × follow-up rate × trial effect. Base: 4.0%. The PS assumed 25%.	Institute_ROI

D · Is it worth it for the institute?

Term	In plain words	Where it shows up
ROI (our definition)	Value gained ÷ cost paid. 2.0x means ₹2 back for every ₹1 paid. This is the PS's definition (₹4,32,000 ÷ ₹1,92,000 = 2.25x). Careful: some textbooks subtract the cost first.	Institute_ROI
Hard-benefit ROI	The same ratio counting only what the owner can bank: fees retained and staff time saved. Base: 0.55x. This is the cautious headline.	Institute_ROI
Fees-only ROI	The same ratio counting only the fees saved. The most cautious version.	Institute_ROI
Soft benefits	Gains that are real but harder to prove: extra admissions because results and parent trust improve (and, to a lesser degree, staff hours saved).	Institute_ROI
Break-even	How many students we must help keep each year so the fees saved equal what the institute pays us. Base: 5.5 students, or 1.5 extra admissions.	Institute_ROI

E · Is it worth it for us?

Term	In plain words	Where it shows up
List vs realised price	List is the brochure price (₹30 per student per month). Realised is what we really collect after discounts (₹25.20).	Pricing
Minimum fee	A floor of ₹3,000 a month per institute so tiny or quiet months do not collapse revenue.	Pricing
Bad debt	Bills that never get paid. Base: 3%.	Unit_Econ
COGS	Cost of goods sold: what it costs to serve a customer. For us: AI, WhatsApp messages, servers. About ₹3.62 per student per month.	COGS_Build

Term	In plain words	Where it shows up
Gross vs contribution margin	Product gross margin (83%) is revenue minus COGS. Contribution margin (78%) also takes off customer-success staff; that is the PS's definition and what peers call gross margin.	Unit_Econ
CAC	Customer acquisition cost: what we spend to win one paying institute (sales time loaded with 30% overhead, travel, marketing, and the trials that do not convert). Base: ₹15,567. **Fully loaded** CAC also adds pilot costs and onboarding: ₹38,392.	Unit_Econ
Pilot / retro-pilot	A free trial. Ours looks backwards: we load last year's Excel and show the owner who left and whether we would have spotted them.	Assumptions → GTM
Onboarding	Setting an institute up: moving its data and training its staff. Base: ₹8,000 for a Mid institute (₹6,400 Small, ₹12,800 Large).	Unit_Econ
Payback	Months until an institute's contribution has repaid what we spent winning it. **Simple** (what the PS and most peers quote): 5.7 months. **Survival-weighted** (allows for institutes that leave, discounted): 6 months.	Unit_Econ
Logo churn	Share of *customers* (institutes, not rupees) that cancel in a month. Base: 3% for monthly payers; 40% are on annual prepay and cannot leave mid-year, so the effective rate is 1.8%.	Unit_Econ
Renewal spike	Extra cancellations when a new session begins: 20% of institutes do not renew. It only applies after a customer has paid 9 months, because nobody renews in their second month.	Unit_Econ
Survival curve	Month-by-month share of institutes still paying. Base annual retention: 64%.	Unit_Econ
Discounting	₹100 received next year is worth less than ₹100 today. We discount at 20% a year.	Unit_Econ
LTV	Lifetime value: all the contribution one institute brings before it leaves (60-month horizon). Discounted: ₹1.40 lakh. Undiscounted (what many peers quote): ₹1.85 lakh.	Unit_Econ
LTV ÷ CAC	Rupees of lifetime contribution for every rupee spent winning the customer. Base: 3.6x discounted, 4.8x undiscounted (PS: 10.7x). A common rule of thumb is that 3x or more is healthy.	Unit_Econ
Channel partner	A reseller (OMR printers, test-series vendors) who brings institutes for a commission (20% of year-1 revenue). For small institutes it beats field sales: LTV ÷ CAC 3.4x instead of 1.5x.	Unit_Econ
NRR	Net revenue retention: how revenue from existing customers changes over time, including upsell. The PS lists it as a concept to know, but our model has no upsell revenue, so it is not computed.	not modelled

F · About the 3-year plan

Term	In plain words	Where it shows up
Cohort	All the institutes that started paying in the same month. We follow each cohort separately because they face renewal at different times.	Projection
Addressable market (SAM)	The number of institutes we could realistically sell to: independent, 150–2,000 students, in five states. Base: 1,180. Derived, not counted.	Assumptions → Projection
Penetration cap	The most of that market a young company can win in three years: 8% Base, so about 94 institutes in total.	Projection
EBITDA	Earnings before interest, tax, depreciation and amortisation. For us, simply revenue minus every cost of running the business.	Projection
Run-rate revenue	Paying institutes at year end × what an average institute pays in a year. A seasonally fair measure of the size of the business.	Projection

Term	In plain words	Where it shows up
Receivables	Money customers owe us. We assume invoices are paid after 30 days.	Projection
Funding ask	The deepest cash deficit before the business pays for itself, plus a buffer of 3 months of costs, rounded up to ₹5 lakh.	Projection

PART 3

The tab-by-tab tour

Thirteen tabs, in the order they appear in the workbook: twelve you can see, plus the hidden Engine (three more hidden sheets hold the fixed-scenario projections). Each gets the same treatment: what it is, why it exists, what you will see, a worked example with real numbers, where the PS asks for it, and what to watch out for.

TAB 1

Cover

the front door

Open the workbook and you land here. It is one page that tells you the answer, tells you whether to trust it, and lets you flip the whole model between three versions of the future.

- **Scenario switch (cell C6).** Type 1, 2 or 3 for Bear, Base or Bull. Every tab recalculates.
- **Status light (cell C7).** Reads *ALL CHECKS PASS* when the arithmetic is consistent, plus a count of honest warnings (currently 3).
- **Headline table.** The key numbers for a typical 400-student institute and for the 3-year plan, ours next to the PS's naive version.
- **Colour code and how-to notes** for first-time users.

Cover headline (Mid institute, Base)	Our model	PS naive
Effective dropout reduction	4.0%	25.0%
Dropouts prevented per year	1.9	12.0
Institute ROI (all benefits)	1.7x	2.25x
Institute ROI (hard benefits: fees + staff time)	0.55x	2.25x
Institute ROI (fees only)	0.35x	2.25x
Break-even: students to save per year	5.5	5.3
Extra admissions needed to break even (after fees + staff time)	1.5	n/a
Platform cost as % of a student's fee	0.38%	0.80%
Avg monthly revenue per institute	₹8,612	₹16,000
Contribution margin	78%	66%
CAC: sales + marketing per paying institute	₹15,567	₹25,000
Fully loaded acquisition cost	₹38,392	₹33,000
LTV (discounted)	₹1.40 lakh	₹3.53 lakh
LTV ÷ acquisition cost (discounted)	3.6x	10.7x
LTV ÷ acquisition cost (undiscounted, peer convention)	4.8x	10.7x
Payback, survival-weighted (months)	6.0	3.1
Payback, simple (months)	5.7	3.1
Institutes needed to cover fixed team cost (steady state)	18.1	n/a
3-year: revenue in year 3	₹65.3 lakh	n/a

Cover headline (Mid institute, Base)	Our model	PS naive
3-year: EBITDA in year 3	₹11.9 lakh	n/a
3-year: paying institutes at the end of year 3	58	n/a
3-year: funding the model says we need	₹45.0 lakh	n/a
3-year: break-even month (3 positive months in a row)	month 16	n/a

FROM THE PS (the case brief)
PS Section 10: *"Teams that clearly state their assumptions and limitations score higher than teams that oversell."* The **Guide (naive)** column and the warnings counter are that principle made visible: we show the PS's version and our own, side by side.

WATCH OUT
The headline follows the switch in C6, so always check it before quoting a number. For slides, quote the Base scenario and say so.

TAB 2

Assumptions

the control panel

Every number the model uses is typed here, and only here. If a number appears inside a calculation tab, it was pulled from this tab. Change a blue cell and the whole workbook updates.

- **102 inputs in eight blocks:** Segments (institute sizes and fees), Institute (dropouts and prevention), Pricing, COGS (cost to serve), GTM (sales and churn), Team, Projection (hiring, market size, cash) and Guide (the PS's own numbers, kept for comparison).
- **Columns:** Bear, Base and Bull (blue, typed) · **LIVE** (yellow: the one the model uses) · Type and Confidence (H, M or L) · Source · Note · **Scenario driver** (Y or N).
- **How LIVE works:** for the 13 inputs marked Y it picks Bear, Base or Bull according to the Cover switch; for every N input it always uses Base. (The Excel function CHOOSE does the picking.)

How much of this is evidence? Of 102 inputs: 23 researched, 1 public, 16 straight from the PS, **57 our own assumptions**, 5 structural. In other words, about half of what drives the model is still an informed guess, and the workbook labels each one so nobody is misled.

The ten inputs that move the answer most (Mid institute):

Input	Bear	Base	Bull	Type	Driver
List price per student per month	₹20	₹30	₹40	Assumption	Y
Extra admissions next session	0%	1%	3%	Assumption	Y
Annual fee per student (Mid)	₹60,000	₹75,000	₹95,000	Researched	N
Dropout reduction from outreach	5%	15%	30%	Researched	Y
Mentor minutes per flagged student	240	120	60	Assumption	N
Flag precision	10%	15%	25%	Researched	Y
Pilot-to-paid conversion	15%	30%	50%	Researched	Y
Monthly churn (monthly payers)	5%	3%	1.5%	Researched	Y
Non-renewal at session end	30%	20%	10%	Assumption	Y
Share of the market won in 3 years	5%	8%	12%	Assumption	Y

FROM THE PS (the case brief)

PS Section 7 gives a "data pack" of baseline values and says *"You may change them with justification."* Rule 4 in Section 11 repeats it. Part 4 shows exactly where we changed the PS's numbers and why.

WATCH OUT

Bear means worse for our business, not "the lowest number". For churn, Bear is 5% and Bull is 1.5%. Always read the column header, not the size of the number. And an input marked N has Bear and Bull values, but the scenario switch ignores them; only the tornado and the Stress test use them.

TAB 3

Seasonality

the calendar of busy and quiet months

Coaching runs on a calendar: students join in April to June, classes thin out around the board and entrance exams, and by March many have finished. A flat "students × price × 12" ignores that. This tab holds three small tables that teach the model the rhythm.

- **A. Three monthly indices (April to March).** How full the classroom is, from 1.00 (full) down to 0.45; the 3 renewal months when institutes decide whether to continue; and a **sales intensity index** that peaks at 2.0 in the weeks before a new session (March to June), which drives how many pilots our reps can start.
- **B. Dropout timing.** Of all students who leave in a session, 3% leave in course month 1 and 43% in months 3 and 4 (after the first big tests and before the second instalment). The first five months hold **71%** of dropouts.
- **C. A 60-month timeline** that maps "our paid month 1, 2, 3…" to calendar months, so Unit_Econ knows which months are quiet and which are renewals.

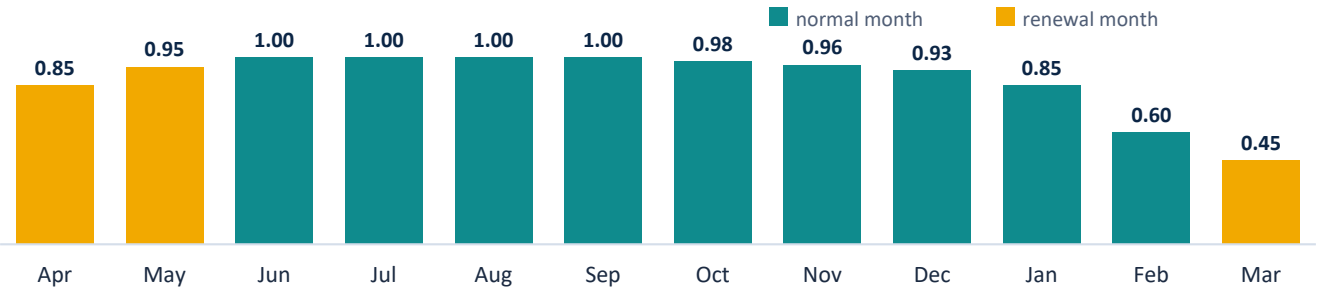


Table A: how full the classroom is, month by month (April to March). Amber = renewal month.

FROM THE PS (the case brief)

PS Section 6 lists "seasonal student counts" as a risk of per-student pricing; Section 7 says dropouts are *"highest in first 4–5 months"* and that the sales cycle *"peaks before new session (March–June)"*. Judges also ask *"what happens to churn after the exam season?"* This tab is the foundation of our answer; the answer itself is in Unit_Econ.

WATCH OUT

Every number on this tab is an assumption (confidence: low). We found no public month-by-month data. The dropout curve follows the PS and the technical team's synthetic data; the rest is shaped by what we did find. All of it must be checked in owner interviews.

TAB 4

Pricing

the price tag after discounts

The brochure says ₹30 per student per month. Nobody pays exactly that. This tab turns the list price into what we really collect.

WORKED EXAMPLE · Mid institute, Base

List price **₹30** per student per month

− negotiation discount 10%, and annual prepay: 40% of institutes take it at 15% off, an average of 6%

= effective discount **16%** → realised price **₹25.20** per student per month (before GST). If the owner cannot reclaim GST, it *feels* like ₹29.74.

A minimum fee of ₹3,000 a month means a small institute still pays something in the quietest month.

The competitor anchor. Owners already pay for management apps. We found public prices of roughly ₹4 to ₹21 per student per month (at 400 students) for tools like Classplus, Fedena and Graphy. Our realised price is **2.4x** the middle anchor (₹10.4). That gap is our biggest commercial risk and the reason the Checks tab shows its first warning.

Packaging in the workbook

Tier	What it contains	Price	Role in the numbers
Starter	CSV or OMR upload, test-level reports, one batch	Free	Lead generation; free pilots are costed
Pro (core)	Topic heatmap, at-risk list with reasons, template parent reports	₹30	The paid plan the whole model is built on
Premium (AI)	GenAI parent reports on WhatsApp, percentile band, study plans	Pro + add-on	**Not** in revenue until priced by research
Annual plan	Pro billed yearly upfront	₹25.50	Cash up front; prepaid institutes cannot churn mid-year

FROM THE PS (the case brief)

PS Section 6 lists seven revenue models. Our numbers are built on the first one, **per-student SaaS** with a minimum fee. "Freemium" appears as the free Starter tier and free pilots. The PS also asks directly: *"Will owners really pay ₹40 per student?"* Part 4 lists which of the seven we model and which we do not.

WATCH OUT

Price pulls two ways. A lower price improves the owner's return (at ₹20: 2.3x) but weakens our LTV ÷ CAC (2.1x). A higher price (₹40) gives 1.4x and 5.2x. The right price is where both work, and only owner interviews can find it.

TAB 5

COGS_Build

the cost of the ingredients, per student

COGS means *cost of goods sold*. For a software company it is what it costs to serve each customer: the AI that writes parent reports, WhatsApp messages, servers, and the paperwork for protecting children's data.

Cost line	How it is worked out	₹ per student per month
AI parent reports	4 reports × (≈1,500 + 1,000 tokens) at small-model prices	₹0.71
WhatsApp messages	6 messages × ₹0.1323	₹0.79
Servers and storage	per-student share of compute, storage, backups	₹0.80
Consent verification	₹1.50 a year ÷ 12	₹0.125
Variable total		**₹2.43**

Cost line	How it is worked out	₹ per student per month
Fixed share	(servers ₹4,000 + WhatsApp provider ₹1,500) ÷ 20 institutes + ₹200 data-protection compliance = ₹475 a month per institute; at 400 students that adds ₹1.19	₹1.19
All-in at 400 students		**₹3.62**

A token is roughly three-quarters of a word; AI providers bill per million tokens.

FROM THE PS (the case brief)

PS Section 3, question **A7**: *"What is your infrastructure + AI cost per student per month?"* Section 7 gives a range of ₹3–10. We land at **₹3.62**, inside it.

WATCH OUT

This models a *paid, hosted* AI service. Our prototype runs the AI on a laptop at ₹0, so real costs could be lower; we chose the cautious route. WhatsApp pricing also changed on 1 October 2026 (replies inside the 24-hour window are no longer free) and the model counts every message.

TAB 6

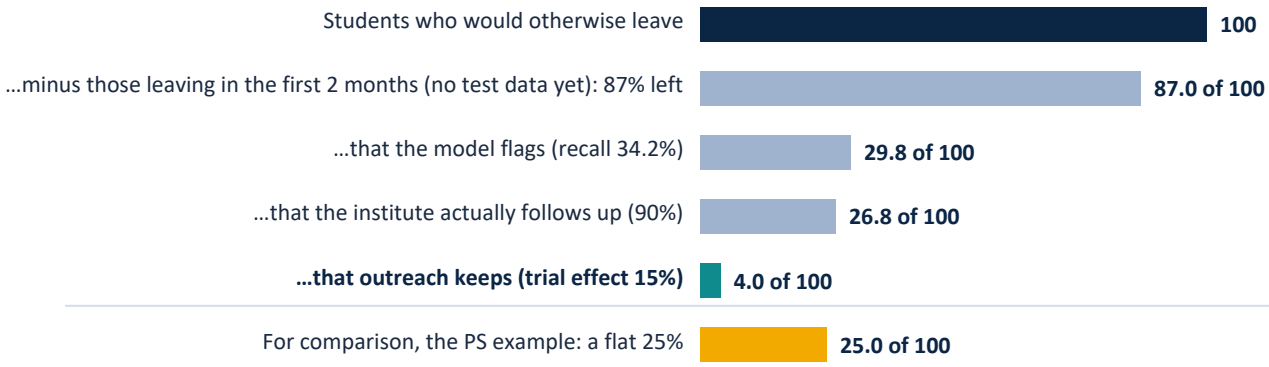
Institute_ROI

the owner's report card

This is the tab an institute owner cares about: *"if I pay for this, am I better off?"* It shows three columns (Small, Mid and Large institutes) and walks down five steps.

1. **What a dropout actually costs.** Not the whole annual fee: only what is still unpaid (or refundable) when the student leaves.
2. **How many we prevent.** A funnel that shrinks the PS's flat 25% into something realistic.
3. **Value gained.** Fees retained, staff time saved, and extra admissions.
4. **Cost paid.** Our fee, GST the owner cannot reclaim, and the mentor time spent acting on flags.
5. **Result.** Three returns (fees only, hard benefits, all benefits) and the break-even.

Step 2 is the heart of it. Of every 100 students who would otherwise leave:

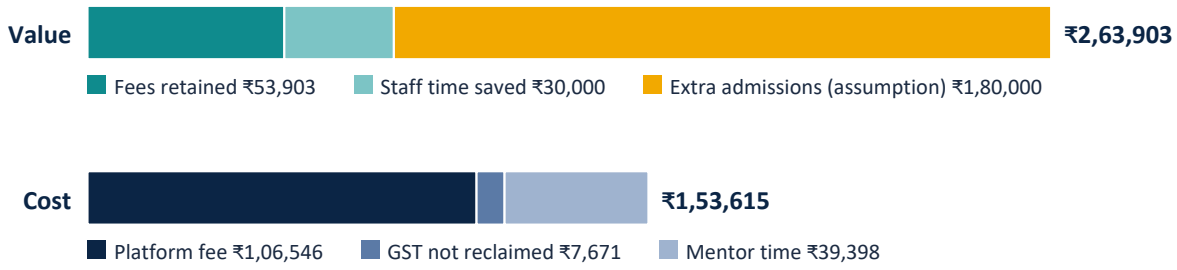


From the PS's 25% to our 4.0%: each bar is what remains after one more real-world filter.

WORKED EXAMPLE · Mid institute (400 students), Base

Line	Number
Students; dropouts a year (12%)	400; 48
Average fee lost per dropout (of a ₹75,000 fee)	₹29,799
Fee lost to dropouts a year, no platform	₹14,30,370
Students we keep (4.0% of 48)	1.9
Fees retained	₹53,903
+ Staff time saved (25 hours a month × ₹100 × 12)	₹30,000
+ Margin on extra admissions (1% × 400 × ₹75,000 × 60%)	₹1,80,000
= Total value	**₹2,63,903**
Platform fee (seasonal students, minimum fee, discounts)	₹1,06,546
+ GST not reclaimed	₹7,671
+ Mentor time acting on flags (27% of the roll flagged, 120 minutes each)	₹39,398
= Total cost	**₹1,53,615**
Return: all benefits · hard benefits · fees only	**1.7x · 0.55x · 0.35x**
Break-even: students kept · or extra admissions	**5.5 · 1.5**

Where the value comes from, and what it costs:



Amber marks the one soft assumption: extra admissions. Without it, value is well below cost.

Results by institute size (Base)

	Small (under 250)	Mid (250–800)	Large (over 800)
Students	180	400	1,200
Return, all benefits	1.6x	1.7x	2.1x
Return, hard benefits (fees + staff time)	0.71x	0.55x	0.53x
Return, fees only	0.28x	0.35x	0.47x
Break-even: students to keep per year	3.1	5.5	12.4

FROM THE PS (the case brief)

This tab rebuilds the PS's own worked example (Section 6) with the same ROI definition, value ÷ cost. The PS assumed 25% of dropouts prevented and asked: *"Will the platform actually prevent 25%… and how would you prove it?"* Two more links:

Section 8 (DPDP Act): the law restricts tracking children, so our recall for under-18s (32%) is lower than for adults (42%).

Section 5 (student well-being): flags must lead to *supportive* actions, so mentor time is counted as a real cost.

WATCH OUT

Retention alone does not pay for the product. Fees retained return 0.35x of the cost; with staff time 0.55x. The owner breaks even if we keep 5.5 students a year (we expect 1.9) **or** if better results bring 1.5 extra admissions a year.

68% of the value is the admissions assumption (1% more admissions), which has no evidence yet. It lifts the return from 0.55x to 1.7x. We show both so nobody is misled. The pitch is a bundle (staff time, parent reports, results), not a retention promise.

TAB 7

Unit_Econ

do we get our money back?

Now we switch sides. This tab follows **one institute** from the day we pitch it to the day it cancels, and asks whether it paid us back. (*Unit economics* is the profit and loss of a single customer.) It has five blocks.

WORKED EXAMPLE · Mid institute, Base

1 · Earn. After seasonality, discounts and 3% bad debt, a Mid institute pays us on average **₹8,612** a month. Minus COGS and support (₹35,000 for a customer-success person ÷ 80 institutes = ₹437.50), contribution is **₹6,740** a month (78% margin).

2 · Win. A rep costs ₹24,000 + ₹5,000 travel a month, loaded with 30% overhead (manager, incentives, recruiting), and starts 10 pilots: ₹3,770 a pilot. Only 30% convert, so ₹12,567 of sales cost per paying institute + ₹3,000 marketing = **CAC ₹15,567**. Add pilot setup (₹3,000), serving the free month (₹1,448), both ÷ conversion, plus onboarding (₹8,000): **fully loaded ₹38,392**.

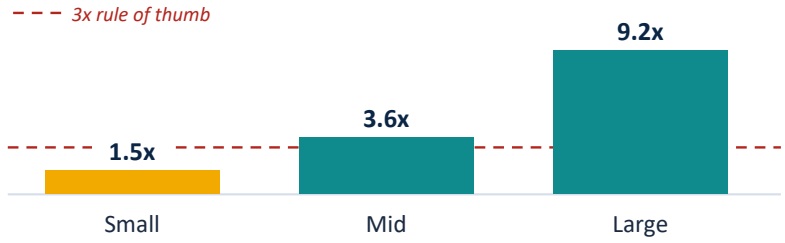
3 · Keep. 40% are prepaid and cannot leave mid-year, so 3% churn applies to the rest: 1.8% a month. Add the renewal spike (20% do not renew, about 7.2% in each of 3 renewal months, and only after 9 months of paying). Annual retention: **64%**; average paid life 27 months.

4 · Add up. Contribution each month × the share still paying × a discount factor, over 60 months: **LTV ₹1.40 lakh** discounted (₹1.85 lakh undiscounted).

5 · Compare. LTV ÷ fully loaded cost = **3.6x** discounted, **4.8x** undiscounted. Payback: **5.7 months** simple, 6 months survival-weighted.

The same walk for each institute size, and field sales versus a channel partner (Base):

	Small	Mid	Large
Average monthly contribution	₹2,569	₹6,740	₹22,044
Fully loaded acquisition cost (field sales)	₹35,009	₹38,392	₹49,677
Lifetime value (discounted)	₹53,353	₹1,39,813	₹4,56,637
LTV ÷ acquisition cost, field sales	**1.5x**	**3.6x**	**9.2x**
Acquisition cost via a channel partner	₹15,794	₹28,670	₹74,810
LTV ÷ acquisition cost, channel partner	**3.4x**	**4.9x**	**6.1x**
Payback (months, survival-weighted)	19	6	3



LTV ÷ acquisition cost by institute size, field sales. Amber is below the 3x rule of thumb.

Break-even scale. The right-hand side of the tab asks: how many institutes cover our own fixed team cost if we stopped selling? Our team is 3 founders at ₹20,000 each, 1 engineer at ₹50,000 and ₹12,000 of tools, which adds up to ₹1,22,000 a month. Divided by one Mid institute's ₹6,740 contribution, that is **about 18 institutes**. The same area also holds the PS's naive numbers. The month-by-month engine rows underneath are grouped and collapsed; click the

"+" in the left margin to expand them.

FROM THE PS (the case brief)

PS Section 6 (platform side): revenue ₹16,000, contribution ₹10,600, payback 3.1 months, LTV/CAC about 10x. Section 8 names the metrics to know: CAC, payback, gross margin, NRR and LTV/CAC. We compute all of them except NRR. Judges ask *"what happens to churn after exam season?"* (block 3) and *"what does it cost to onboard a low-tech institute?"* (block 2 and the size multipliers). Section 3 (B3) asks *"direct sales, channel partners or product-led?"*: the channel rows answer it.

WATCH OUT

LTV is contribution only. It is before the founders' and engineer's salaries, which is why the break-even-scale block and the Projection tab exist.

Support cost is our softest cost number. ₹438 per institute per month is far below the PS's ₹3,000. It rests on one assumption: a customer-success person handling 80 institutes. In Bear (40 institutes, ₹45,000) it is ₹1,125.

Small institutes earn only 1.5x by field sales. A rep is not worth sending; a channel partner takes it to 3.4x.

TAB 8

Projection

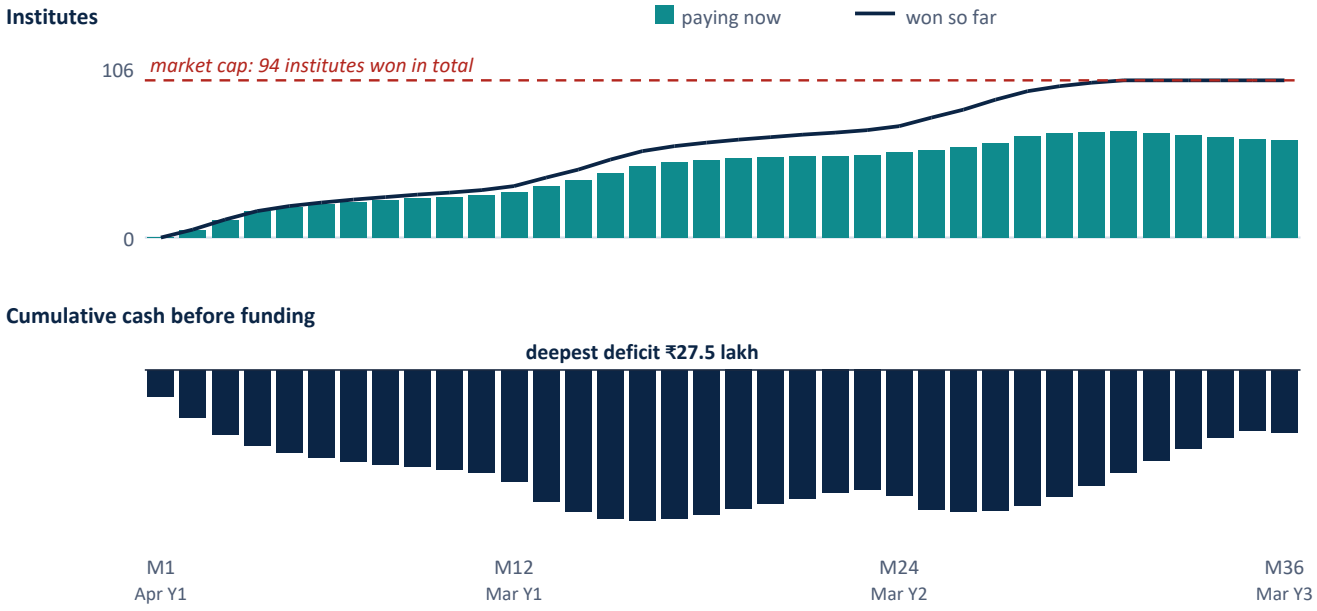
three years, month by month

The PS rubric lists a **3-year projection** under revenue model and unit economics. This tab turns the unit economics into a monthly plan for 36 months (April to March): how many pilots our sales team can run, how many turn into paying institutes (capped by the size of the market), how many stay, what we earn and spend, and how much cash we need.

1. **Sales capacity.** Reps × pilots per rep per month × the sales intensity index (high March to June).
2. **New paying institutes.** Pilots from a month earlier × conversion, but never more than the market cap (addressable institutes × maximum penetration = 94 institutes won in total).
3. **Cohorts.** Each month's new institutes are followed separately with the same churn and renewal logic as Unit_Econ.
4. **Revenue.** Paying institutes × what an average institute pays that month (a weighted mix of Small, Mid and Large).
5. **Costs.** Cost of serving (variable cost, server stacks, customer-success staff), sales and marketing, engineers, and founders and overheads, with floors so costs grow with revenue.
6. **Cash.** Receivables, equipment for new hires, then the funding ask: the deepest cash deficit plus three months of costs.

Base case, by financial year

	Year 1	Year 2	Year 3	3 years
Revenue	₹18,87,784	₹48,02,825	₹65,28,324	₹1,32,18,933
Gross profit (after support)	₹11,50,872	₹35,26,610	₹49,32,011	₹96,09,494
Sales and marketing	₹12,99,148	₹13,91,440	₹13,37,028	₹40,27,617
Product and engineering	₹6,00,000	₹12,96,000	₹13,99,680	₹32,95,680
Founders and general costs	₹8,64,000	₹9,33,120	₹10,07,770	₹28,04,890
EBITDA	**-₹16,12,276**	**-₹93,950**	**₹11,87,534**	**-₹5,18,692**
New paying institutes	31	36	28	94
Paying institutes at year end	28	51	58	
Headcount at year end	6	7	7	



Paying institutes (bars), institutes won so far (line) and the market cap (dashed), above; cash before any funding, below.

WORKED EXAMPLE · Base

Funding the model says we need: ₹45,00,000 (₹45.0 lakh). The deepest cash deficit is ₹27.5 lakh, plus a buffer of ₹13.1 lakh (three months of costs at that point), rounded up to ₹5 lakh.

Break-even: the first three months in a row with positive EBITDA is month 16; on a trailing-12-month basis, which ignores the seasonal dip, it is month 25.

Market: at month 36 we serve 4.9% of the addressable market. The cap limited sales in 6 months, so a second sales rep would not help; new territory would.

Realised CAC: sales and marketing spend ÷ new institutes is ₹42,665, above the unit-economics ₹38,392, because a rep is paid in the quiet months too.

FROM THE PS (the case brief)

PS Section 4 and Section 10 name the *3-year projection* under "Revenue model & unit economics", and Section 9 asks for a "detailed financial model (3-year projection, scenarios, sensitivities)" in Round 2. Section 7 says the sales cycle *"peaks before new session (March–June)"*, which is the sales intensity index. Section 3 (B3) asks which cities to enter first: the market cap is built for the PS's cities (Patna, Lucknow, Kota, Jaipur, Indore).

WATCH OUT

The market size is derived, not counted. We found no institute counts by size for any state (research R7), so the 1,180 addressable institutes and the 8% penetration are assumptions. They are the biggest lever on the 3-year result.

Bear funding assumes no cost response. The hiring plan is fixed, so Bear shows what happens if we keep the team and the market disappoints. A real team would cut costs.

Not in the cash view: income tax, early cash from annual prepay (which would help), and salary step-ups beyond inflation. The plan has one field rep because at this market size one rep exhausts the market in about three years.

TAB 9

Scenarios

three weather forecasts and a storm

Rather than one answer, this tab shows three side by side: **Bear** (the key drivers go badly), **Base** (our best guess) and **Bull** (they go well), plus the PS's naive column for contrast. Only the 13 scenario drivers move; the rest stay at Base. A fourth column, **Stress**, puts every input at its worst at once. They are always shown together, whatever the Cover switch says.

Mid institute	PS naive	Bear	Base	Bull	Stress
Share of dropouts prevented	25.0%	1.0%	4.0%	11.3%	0.6%
Students kept per year	12.0	0.3	1.9	8.2	0.2
Owner's return, all benefits	2.25x	0.37x	1.7x	3.9x	0.09x
Owner's return, hard benefits	2.25x	0.37x	0.55x	1.3x	0.09x
Owner's return, fees only	2.25x	0.09x	0.35x	1.1x	0.02x
Break-even students per year	5.3	3.8	5.5	7.2	11.7
Monthly contribution per institute	₹10,600	₹3,903	₹6,740	₹9,576	-₹1,560
CAC (sales + marketing)	₹25,000	₹44,889	₹15,567	₹8,386	₹65,111
Fully loaded acquisition cost	₹33,000	₹82,540	₹38,392	₹25,281	₹1,90,892
Annual institute retention	69.4%	48.6%	64.3%	80.7%	42.9%
Lifetime value (discounted)	₹3,53,333	₹61,848	₹1,39,813	₹2,65,113	-₹20,910
LTV ÷ acquisition cost, discounted	10.7x	0.7x	3.6x	10.5x	-0.1x
LTV ÷ acquisition cost, undiscounted	10.7x	0.9x	4.8x	14.8x	-0.1x
Payback, simple (months)	3.1	21.1	5.7	2.6	never

By institute size, the owner's return and our LTV ÷ acquisition cost:

	Return Bear	Return Base	Return Bull	LTV/CAC Bear	LTV/CAC Base	LTV/CAC Bull
Small	0.67x	1.6x	3.4x	0.28x	1.5x	4.7x
Mid	0.37x	1.7x	3.9x	0.75x	3.6x	10.5x
Large	0.21x	2.1x	5.1x	2.1x	9.2x	24.9x

The 3-year plan under each scenario:

	Bear	Base	Bull
Revenue in year 3	₹11.0 lakh	₹65.3 lakh	₹2.50 crore
EBITDA in year 3	-₹30.2 lakh	₹11.9 lakh	₹1.26 crore
Paying institutes at the end of year 3	15	58	191
Funding the model says we need	₹1.00 crore	₹45.0 lakh	₹30.0 lakh
Break-even month (3 positive months in a row)	after month 36	month 16	month 4

> **FROM THE PS (the case brief)**
> PS Section 6: *"Build a model with ranges, scenarios and evidence."* Round 2 (Section 9) asks for a "detailed financial model (3-year projection, scenarios, sensitivities)."

WATCH OUT

Bear and Bull are coherent stories, not extremes: price, trial effect, admissions gain, recall, precision, conversion, churn and market size all move together. **Stress is not a forecast**: with every input at its worst, contribution turns negative (-₹1,560 a month). Bear's funding need (₹1.00 crore) assumes the team stays the same size while the market disappoints.

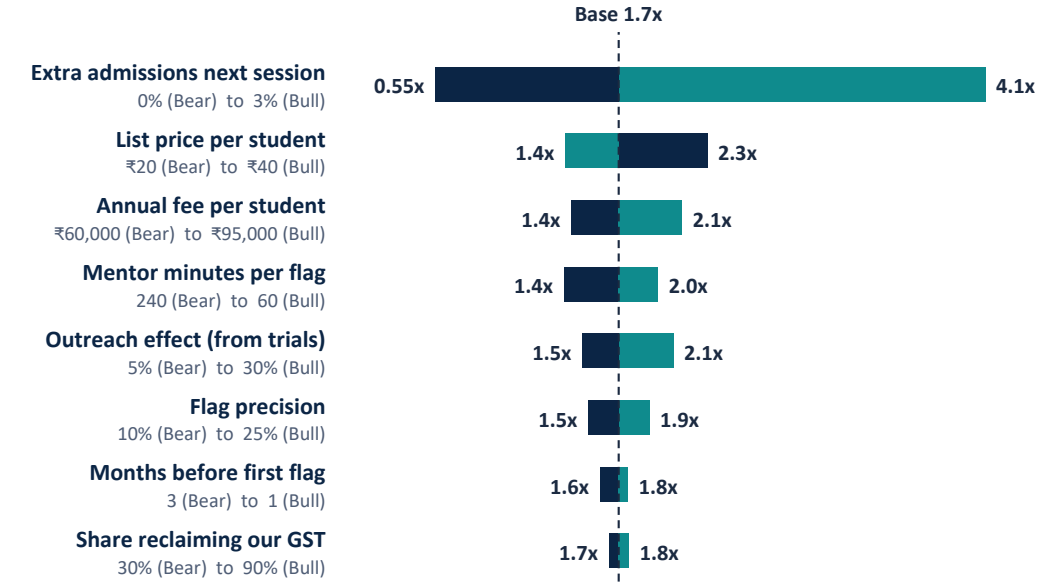
TAB 10

Sensitivity

which dial matters most?

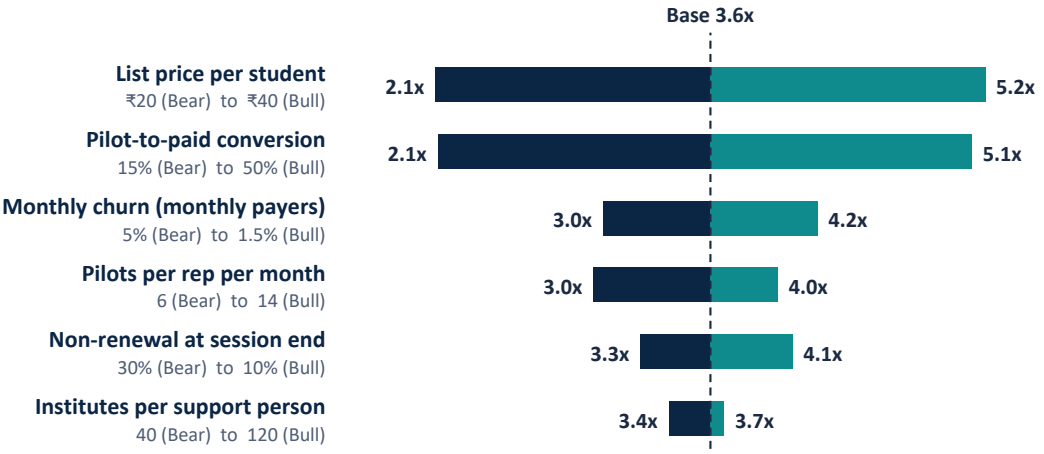
Scenarios move several things together. This tab moves **one thing at a time** to find the dials that really matter. It has a tornado chart and two what-if grids.

Owner's return, all benefits: each input moved from its Bear value to its Bull value, everything else at Base



Navy end = input at its Bear value; teal end = input at its Bull value. A long bar means the input matters; a short bar means relax.

Our LTV ÷ acquisition cost, same method



Inputs that only affect the owner's return (like the dropout rate) do not move this chart.

What the tornado tells us

- **For the owner's return**, the biggest dial by far is extra admissions (0.55x to 4.1x), then list price (2.3x at ₹20 to 1.4x at ₹40), annual fee (1.4x to 2.1x) and mentor minutes per flag (1.4x to 2.0x).
- **For our LTV ÷ CAC**, the big ones are list price (2.1x to 5.2x), pilot conversion (2.1x to 5.1x) and churn and renewal (3.0x to 4.2x; 3.3x to 4.1x).
- **The trial effect** (1.5x to 2.1x) and **flag precision** (1.5x to 1.9x) matter, but much less than whether the owner believes in the soft benefits.

Grid 1: owner's return by list price and effective dropout reduction

Reduction ↓ / price →	₹20	₹30	₹40	₹50	₹60	₹80
5%	2.4x	1.8x	1.4x	1.2x	1.0x	0.81x
8%	2.7x	2.1x	1.7x	1.4x	1.2x	0.92x
11%	3.1x	2.3x	1.9x	1.6x	1.3x	1.0x
15%	3.6x	2.7x	2.1x	1.8x	1.5x	1.2x
20%	4.1x	3.1x	2.5x	2.1x	1.8x	1.4x
25%	4.7x	3.6x	2.8x	2.4x	2.0x	1.6x

Red: under 1x · amber: 1–2x · green: 2x or more. Includes the 1% admissions assumption.

Grid 2: our LTV ÷ acquisition cost by monthly churn and acquisition cost

Cost ↓ / churn →	1%	2%	3%	4%	5%
₹15,000	11.5x	10.3x	9.3x	8.5x	7.8x
₹25,000	6.9x	6.2x	5.6x	5.1x	4.7x
₹35,000	4.9x	4.4x	4.0x	3.6x	3.3x
₹50,000	3.4x	3.1x	2.8x	2.5x	2.3x
₹75,000	2.3x	2.1x	1.9x	1.7x	1.6x

Red: under 3x · green: 3x or more. Our own acquisition cost is about ₹38,000; at that cost 3x needs churn below 4%.

FROM THE PS (the case brief)

PS Section 9 requires *"at least one sensitivity analysis"* in Round 1, and Section 10 lists "sensitivity" under unit economics.

WATCH OUT

Labels can confuse. For price, **Bear (₹20) is bad for us but good for the owner's return**. Read the numbers, not the label.

TAB 11

Checks

the smoke alarm

A model can add up correctly and still be wrong in logic. This tab runs **23 automatic error checks** (any failure turns red and the Cover says so) plus **12 warnings**.

- **Error checks** test things that must always be true: the dropout timing adds to 100%; every percentage input sits between 0% and 100%; the customer mix adds to 100%; each Live value lies between its Bear and Bull values.
- **Consistency checks** prove the hidden Engine reproduces the visible return and LTV for all three institute sizes, and that the live Projection equals the fixed-scenario sheet for the scenario you chose.
- **Projection checks** confirm we never have more paying institutes than we have won, never exceed the market cap, and never run out of cash once the funding ask arrives.
- **One reproduction check** confirms our copy of the PS example gives 2.25x, proving we read the guide correctly.
- **Warnings** are honest red flags, not bugs. They fire when the model is telling you something uncomfortable.

Warning	Status now
RCT effect exceeds share of dropouts that are preventable	clear
List price above the highest ERP/LMS anchor found (₹21/student/month)	**FIRING**
Months with negative contribution (Mid)	clear
All-in COGS outside tech band ₹3–10 per student-month	clear
Mid institute ROI (all benefits) below 1.0x	clear
Mid hard-benefit ROI (fees + staff time) below 1.0x	**FIRING**

Warning	Status now
More than 30% of the roll flagged per year (alert fatigue)	clear
Mid LTV/CAC (discounted) below 3x	clear
Mid payback above 24 months	clear
3-year projection never reaches 3 positive EBITDA months in a row	clear
EBITDA in year 3 is negative	clear
The market cap limited sales in the live projection	**FIRING**

FROM THE PS (the case brief)
PS Section 10: *"Teams that clearly state their assumptions and limitations score higher."* The warnings are limitations the model states about itself.

WATCH OUT
Passing checks means the arithmetic is internally consistent. It does **not** mean the assumptions are true.

TAB 12

Sources

the receipts

Every input is listed again with where it came from: its type, a confidence level (H, M or L), the source and a note. If a judge asks *"where did ₹75,000 come from?"*, this is the tab you open.

Evidence type	Inputs	Meaning
Researched	23	found in a named source (a pricing page, a study, a statute)
Public	1	from a public dataset
Guide	16	straight from the PS; kept for comparison
Assumption	57	our own judgement; the number to challenge
Structural	5	a modelling convention, such as the discount rate

The raw research behind the "Researched" inputs is saved in the `research` folder: R1 competitor prices, R6 unit costs (WhatsApp, AI, GST, payments, cloud), R2 fees and refund rules, R3 dropout evidence, R4 trials on early warning and outreach, R5 software-business benchmarks and salaries, R7 market size. Several inputs also cite the technical team's own measurements (recall, precision, minors share, dropout timing).

FROM THE PS (the case brief)
PS Section 8: *"Always cite what you use."* Rule 3 (Section 11) asks for citations for public data and Rule 4 asks for evidence behind any changed assumption.

WATCH OUT
Some sources are weak (blogs and aggregator pages) and are marked confidence L. The dropout rate rests on the PS baseline and a trade-body quote ("at least 10%"), and the market size on a derivation. We show this rather than hide it.

TAB 13

Engine

the hidden backstage

There is one more tab, hidden on purpose, and three hidden projection sheets beside it. The Engine holds 54 copies of the whole calculation chain, one for each case that Scenarios and Sensitivity need: three institute sizes under Bear, Base and Bull, the Stress test, every tornado bar, and the churn grid. The three Proj_ sheets hold the 3-year projection for Bear, Base and Bull. They exist so those tabs work without macros or Excel's Data Table feature, which breaks in Google Sheets. The Checks tab proves they agree with the visible tabs. You never need to touch them.

The files behind the workbook

File	What it does
assumptions.csv, seasonality.csv	The source of every input. Edit these for permanent changes.
build_model.py, proj_sheet.py	Rebuild the workbook from the CSVs: python build_model.py
verify_model.py	Recalculates without Excel and prints headline numbers and checks: python verify_model.py --all
MODEL_AUDIT.md	The CFO review: what was wrong in v1.0, what changed, what is still weak
README.md, DECK_NUMBER_UPDATES.md	Slide-ready numbers, the deck story, a “do not claim” list and old → new numbers
research/	The raw research notes, with links

IN PLAIN ENGLISH

Two ways to change a number. To play, edit a blue cell in Excel. To make it permanent, edit the CSV and rebuild; otherwise the workbook and the CSV drift apart.

PART 4

Linking the model to the PS

The case brief is the exam paper; the model is our answer sheet. Here is where each question is answered, how the judges' four challenges are met, which revenue models are in the numbers, and where we deliberately departed from the PS's data pack.

4.1 Which PS question is answered where

PS ref	What it asks	Where we answer
A7 · Architecture	Infrastructure + AI cost per student per month	COGS_Build: ₹3.62, inside the PS range of ₹3–10
A8 · Privacy and safety	Minors, consent, well-being	Lower recall for under-18s; consent and compliance costs in COGS_Build; mentor time as a supportive cost
B1 · Customer and value	Who buys; the measurable ROI for an institute	Institute_ROI (three returns, break-even); Pricing (what the owner feels)
B2 · Pricing and unit economics	What we charge; CAC, payback, churn, LTV	Pricing, COGS_Build, Unit_Econ
B3 · Go-to-market	Which cities and sizes first; direct, partners or product-led	Unit_Econ by size and channel (Small 1.5x by field sales, 3.4x by partner); Projection market cap built for the PS's cities
B4 · Competition and moat	How we beat generic apps; why our edge grows	Pricing competitor anchor (₹4–21). The model gives the data moat **no credit** (no improving accuracy over time), which is deliberately conservative

PS ref	What it asks	Where we answer
Section 4 and 10 · Judging	Revenue model and unit economics 20% (incl. 3-year projection); Market and GTM 10%	Pricing, Unit_Econ, Projection, Scenarios, Sensitivity; segment results and competitor anchor

4.2 The four questions the PS says judges will ask (Section 6)

Judge's question	Our honest answer	Look at
Will owners really pay ₹40 per student?	We model a ₹30 list price (₹25.20 realised), above the ₹4–21 competitor anchors. It costs the owner about ₹286 a student a year (0.38% of the fee), and returns 0.55x on hard benefits. **Unproven until owner interviews.**	Pricing, Sensitivity
Will you really prevent 25% of dropouts, and how would you prove it?	No. We model 4.0%, built from 8 trials in other settings and the technical team's measured recall and precision. Proof would come from a staggered rollout across batches in a pilot.	Institute_ROI
What happens to churn after the exam season?	20% of institutes do not renew at session end (after nine months of paying); prepaid ones cannot leave mid-year. Annual retention is 64%.	Seasonality, Unit_Econ
What does it cost to onboard a low-tech institute?	₹8,000 for a Mid institute (₹6,400 Small, ₹12,800 Large) after a light ₹3,000 retro-pilot; ₹38,392 fully loaded. The Bear case assumes ₹15,000.	Assumptions, Unit_Econ

4.3 The PS's seven revenue models: which are in the numbers

Model (PS Section 6)	In our numbers?	Note
Per-student SaaS	**Yes, the core**	List price × active students, with a minimum fee
Tiered plans	Described, not priced	Only the Pro tier carries revenue
Freemium + AI add-ons	Partly	Free Starter tier and free pilots are costed; the AI add-on is not priced
Outcome-linked pricing	No	A natural answer to the hard-benefit gap, but attribution disputes make it hard to measure; a pilot-only option for Round 2
Parent subscription (B2B2C)	No	Needs the institute's agreement; no evidence yet
Marketplace / upsell	No	Would distract from the core
Aggregated benchmarks	No	Strict privacy limits; individual data must never be exposed

4.4 Where we changed the PS's data pack, and why

PS Rule 4 (Section 11) says *"You may change any assumption if you justify it with evidence."* Here is every major change, so no one can say we hid it:

Item	PS	Ours (Base)	Why
Annual fee per student	₹60,000	₹75,000	Local Tier-2 fee pages; the PS's ₹60k is our Bear
Price per student per month	₹40	₹30 list; ₹25.20 realised	Competitor prices are ₹4–21; discounts are real
Dropout rate	12%	12%	Same as the PS; the technical team's synthetic data uses 18% to get enough events
Dropouts prevented	25%	4.0%	8 trials × measured recall and precision × month-1 leavers

Item	PS	Ours (Base)	Why
Fee lost per dropout	₹36,000	₹29,799	Unpaid instalments and refunds, by month of leaving
Infra + AI per student per month	₹6	₹3.62	Built bottom-up; inside the PS's ₹3–10
Support per institute per month	₹3,000	₹438	One CSM per 80 institutes (our softest cost number)
CAC + onboarding	₹25,000 + ₹8,000	₹15,567 CAC; ₹38,392 fully loaded	Rep cost with 30% overhead, pilots that do not convert, onboarding
Monthly churn	3% flat	1.8% + 20% at renewal	Prepaid cannot leave mid-year; renewal is the real risk
Payback	3.1 months	5.7 months simple	Higher acquisition cost, seasonal revenue
LTV ÷ CAC	10.7x	3.6x discounted; 4.8x undiscounted	The result of all the above

PART 5

Try it yourself in five minutes

Reading about a model only gets you so far. Open the workbook and follow these steps.

1. **Open the Cover tab.** Check that cell C7 says *ALL CHECKS PASS*.
2. **Flip the switch.** Type 1 in cell C6 and watch every headline number change (Bear). Type 3 for Bull, then return to 2.
3. **Change one belief.** Go to Assumptions, find a blue Base cell, change it, and look at the Cover again.
4. **Read the tornado.** On Sensitivity, find the inputs with the longest bars. Those are the ones worth a conversation with an owner.
5. **Follow the money.** On Projection, look at the cash chart and the funding ask.
6. **Check the receipts.** On Sources, look up any input you doubt.

Three experiments to try

Do this (Assumptions, Base column)	You should see (Mid institute)
Set *Extra admissions next session* from 1% to 0%	Owner's return, all benefits, falls from 1.7x to 0.55x: the soft benefit was carrying it.
Raise *List price* from ₹30 to ₹40	Owner's return falls to 1.4x, but our LTV ÷ CAC rises to 5.2x.
Set the Cover switch (C6) to 1 (Bear)	Owner's return drops to 0.37x and LTV ÷ CAC to 0.75x; year-3 revenue falls to ₹11.0 lakh.

Before you quote a number on a slide

- Check the Cover says *ALL CHECKS PASS* and the switch is on 2 (Base).
- Label it **Model output**, never “measured”. State the evidence type of the inputs behind it (Part 3, Assumptions).
- Do not claim “we reduce dropouts by X%”. Say: *trials in comparable settings show 5–30% among students who get outreach; we model 15%.*
- Pair the all-benefit return with the hard-benefit return, so no one can say the soft benefits are hiding the truth.

PART 6

Honest limits and what is next

The PS rewards teams that state their limits before the judges do. Here are ours.

What changed after the CFO review

The first version of this model was reviewed line by line against the PS, the technical team's measurements and how SaaS peers calculate. The full record is in `MODEL_AUDIT.md`. The big changes:

Change	Why
Recall, precision, minors share and dropout timing now match what the technical team measured	We had assumed a model twice as good as the one being built
Mentor time per flagged student raised to 2 hours	The trials behind the 15% effect used repeated outreach
A renewal decision only exists after 9 months of paying	A May sign-up was facing renewal in June
CAC loaded with sales overhead; simple payback and undiscounted LTV added	Peers and the PS quote these
Hard-benefit return now leads; admissions shown separately	One unproven assumption carried two-thirds of the value
Bear / Base / Bull flex 13 drivers; a separate Stress test	Everything-at-worst is not a scenario
New Projection tab with a market cap and a funding ask	The PS rubric names a 3-year projection

The weakest numbers

Input	Why it is weak	How we firm it up
List price (₹30)	Willingness to pay is untested; it is above every competitor price found	Owner interviews with a Van Westendorp price survey
Extra admissions (1%)	No evidence; lifts the owner's return from 0.55x to 1.7x	Track next-session admissions in the first pilots
Addressable market (1,180) and penetration (8%)	Derived: no institute counts by size exist; it caps the whole 3-year plan	10–15 owner calls; the UP district survey when published
Pilot-to-paid conversion (30%)	Borrowed from software free trials, not assisted pilots	Record the first 20 pilots
Recall and precision	Placeholders matching the tech team's early tests	Replace with metrics.json from the technical team
Mentor time (120 min a flag)	No data on how much outreach the trial effect needs	Time-track mentors in pilots
Support per institute (₹438)	Rests on 80 institutes per customer-success person	Time-track onboarding in pilots
Dropout rate and timing, upfront share, refund compliance	One trade-body quote; the rest are assumptions	Ask owners for last session's leavers, by month
GST-registered share (60%)	No data on how many institutes can reclaim GST	Ask in interviews; confirm with a CA

What is not in the model

- **Upsell or expansion revenue**, so no NRR.
- **Outcome-linked pricing** and the other PS revenue models (see 4.3).
- Income tax, early cash from annual prepay, and any **management response** in Bear.
- A **3-year sensitivity** grid (price × conversion → year-3 EBITDA); the tornado covers the owner's return and unit economics only.

- Any benefit from the **data moat**.

> **IN PLAIN ENGLISH**
>
> **The honest one-liner.** On our own numbers, retention alone does not pay for the product: a Mid institute gets back 0.55x of its cost on hard benefits and 1.7x if admissions improve. For us, each institute is worth about 3.6x what it costs to win. In three years that is a lean, market-capped business (₹65.3 lakh revenue, break-even in month 16, ₹45.0 lakh of funding). Promising and *unproven*. The next job is evidence: owner interviews and a pilot.

Next steps

1. Run owner interviews and the price survey; update `assumptions.csv`.
2. Replace recall and precision with the technical team's measured values.
3. Rebuild and verify the workbook, then refresh this guide and the deck numbers.
4. Round 2: a 3-year sensitivity, an outcome-linked pricing option, expansion revenue and NRR.

APPENDIX

Cheat sheet and formulas in plain words

A · One line per tab

Tab	One line	Look here first
Cover	Headline answer, scenario switch, status light	Cell C6 (switch) and C7 (checks)
Assumptions	Every input, with Bear / Base / Bull, source, confidence and driver flag	The yellow LIVE column
Seasonality	Busy and quiet months; when students leave; when we sell	Tables A and B
Pricing	From list price to price actually collected	Realised price
COGS_Build	Cost to serve one student per month	All-in COGS
Institute_ROI	Is it worth it for the owner?	The three returns and break-even
Unit_Econ	Do we earn our money back?	LTV ÷ CAC and payback
Projection	Three years: customers, P&L, cash, funding ask	The annual table and the funding line
Scenarios	Bear, Base, Bull and Stress side by side, plus the PS column and the 3-year plan	Mid-institute table
Sensitivity	Which input moves the answer most	The tornado
Checks	Automatic tests and honest warnings	ERRORS and WARNINGS totals
Sources	Where every input came from	Type and Confidence columns
Engine, Proj_* (hidden)	Backstage copies feeding Scenarios and Sensitivity	Nothing: do not edit

B · The main formulas, in words

What it works out	The formula in words	Mid institute, Base
Price actually collected	list price × (1 − negotiation − prepay uptake × prepay discount)	₹30 × (1 − 16%) = ₹25.20
Share of dropouts prevented	catchable share × recall × follow-up rate × trial effect	87% × 34.2% × 90% × 15% = 4.0%
Owner's return, all benefits	(fees retained + staff time + admissions margin) ÷ (platform fee + GST not reclaimed + mentor time)	₹2,63,903 ÷ ₹1,53,615 = 1.7x

What it works out	The formula in words	Mid institute, Base
Owner's return, hard benefits	(fees retained + staff time) ÷ the same cost	₹83,903 ÷ ₹1,53,615 = 0.55x
Share of institutes still paying	last month's share × (1 − monthly churn) × (1 − renewal spike if it was a renewal month and the customer has paid 9+ months)	annual retention 64%
Lifetime value	sum over 60 months of: share still paying × contribution × discount factor (omit the factor for undiscounted)	₹1,39,813 discounted; ₹1,84,576 undiscounted
Payback	simple: acquisition cost ÷ average monthly contribution; survival-weighted: first month where cumulative discounted contribution ≥ acquisition cost	5.7 months; month 6
LTV ÷ CAC	lifetime value ÷ fully loaded acquisition cost	3.6x discounted; 4.8x undiscounted
New paying institutes in a month	the smaller of (pilots from a month earlier × conversion) and (market cap − institutes won so far)	cap = 1,180 × 8% = 94
EBITDA in a month	revenue − cost of serving − sales and marketing − engineers − founders and overheads	year 3: ₹11.9 lakh
Funding the model says we need	deepest cumulative cash deficit + 3 months of costs at that point, rounded up to ₹5 lakh	₹27.5 lakh + ₹13.1 lakh → ₹45.0 lakh

This guide explains model v1.1 CFO audit · 2026-10-22 and is generated from the workbook by the scripts in the guide folder. It describes assumptions and model outputs; it is not financial or legal advice.